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RESEARCH PAPER

Where does it come from and where does it go? Fast, open-source flow tracing for MODFLOW 6 models

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Abstract

Groundwater studies often ask where water comes from and where it goes. Which land feeds a well, spring or stream? How much of a well's water is river water? Where does recharge under a polluted field end up? We present open-source code, `gwsource`, that answers such questions from the flows a MODFLOW 6 model saves. The code treats the water in each cell as fully mixed, like earlier methods. It computes the share of each cell's water that ends at a well, a stream or another outlet. The same trace splits a well's water by source and gives mean travel times. The code extends the method to every MODFLOW 6 grid type, stream network and linked model. It handles weak sinks. It carries tagged water through storage between time steps, forward and backward in time. Every result is checked against the flow model's water budget. The paper presents ten worked examples. Each checks an output a practitioner would use against other commonly used approaches. Results matched MODFLOW 6 transport to its solver tolerance with the same water volumes. They matched particle tracking in totals and mean travel times, as any method that conserves water should. At the edges of areas the two differ cell by cell. On a regional model of 1.68 million cells, one trace of eight wells, alone and together, took 2–3 s. That is small next to the model run itself. Tracing is cheap enough to include in history matching and similar analyses. Full-mixing areas are wider than particle-tracked areas. Particle tracking remains better where a sharp edge or first arrival matters.

KEYWORDS
flow tracing, full mixing, MODFLOW 6, contributing area, sources of water, travel time, transient tracking, ensemble, source protection

ARTICLE IMPACT STATEMENT

Open-source code turns saved MODFLOW 6 flows into contributing areas, water sources and travel times for any well or stream, in steady-state or transient flow.

1. INTRODUCTION

Practitioners who ask where groundwater comes from and where it goes also need to know how the answers change through the year and how sure they can be of them.

Methods that answer these questions from a groundwater flow model fall into four families. Particle tracking follows particles of water along flow paths. It is the standard tool for mapping a well's contributing area. This is the land where water enters the aquifer and later reaches the well. In MODFLOW models it runs in MODPATH (Pollock 2016) or in the particle-tracking model of MODFLOW 6 (PRT; Langevin et al. 2026). Transport with tagged water gives the water from one source a concentration of 1 and all other water a concentration of 0. A cell's concentration is then the share of its water that came from that source (Bedekar et al. 2024). In MODFLOW 6 it runs in the Groundwater Transport (GWT) Model, with one model for each source (Langevin et al. 2022). As documented, the unit-concentration method steps forward in time only. It gives the share of each cell's water that came from a source, but not the share that will later reach a well. Adjoint and age methods solve transport equations of their own (Neupauer and Wilson 1999, Goode 1996). Full-mixing flow tracing splits each cell's mixed water among its outflows.

Particle tracking has one difficulty at weak sinks. Sink terms are the boundaries that remove water from a cell, such as a pumping well or a gaining stream. A weak sink is a cell whose sink terms take only part of the cell's outflow. A particle cannot split. It must either stop at a weak sink or pass on. If it stops, all

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its water is counted as reaching the sink. If it passes, none of it is. The rule chosen changes the contributing area (Section 5).

Under full mixing, a weak sink can take just its share of the cell's water. The rest passes on to the neighbouring cells. Two published full-mixing programs apply different rules at a weak-sink well. FlowSource (Foley and Black 2013) is used to draw source protection zones in England (Environment Agency 2019). These zones restrict activities that could pollute the water a supply well pumps. FlowSource counts all the water passing through a weak-sink well's cell as the well's water. We refer to this as net accounting. For transient models, it traces each time step as a steady-state flow field. We refer to such a trace as a snapshot. MODALL (Potter et al. 2008) applies full mixing to steady-state MODFLOW-88 models. At a weak-sink well it counts only the water the well pumps. We refer to this as gross accounting. All these methods are compared in Supporting Information, Table S1.

This paper presents open-source code, `gwsource`, that extends the full-mixing method of FlowSource to MODFLOW 6. It reads every MODFLOW 6 grid type, stream networks and linked models. It uses gross accounting by default. It checks every result against the water budget. In transient models, it carries tagged water through storage, forward and backward in time. Risk-based studies run an ensemble of model versions that fit the measurements about equally well (Fienen et al. 2022). The code is fast enough to trace every version.

Section 4 works through ten examples. Each checks an output a practitioner would use against another approach. Section 5 applies the code to a published regional model. Section 6 sets out the trade-offs of full mixing.

2. METHOD

Overview. This section explains how the code traces water and what each of its outputs means. A backward trace follows the water of a destination, such as a well, back to where it entered the model. A forward trace follows the water of a source, such as recharge, to where it leaves. Both work on the cell-by-cell flows that a MODFLOW 6 model saves, for one time step or for every step of a transient model. The backward fraction F maps where a sink's water comes from. A contributing area follows from it (example A in Section 4). The forward fraction G maps where a source's water goes. A trace also splits a sink's water by source, for example into recharge and river water. With a porosity, a trace also gives mean travel times (example C). For transient models, the code either traces each time step on its own flows or carries tagged water through storage (example B). The derivations and the solution of the equations are given in the Supporting Information (Supporting Information, Text S1).

Full mixing and the cell budget. Every output of the code rests on the assumption of full mixing. Full mixing treats each active model cell as one well-mixed volume of water. All the water leaving a cell then has the same mix of origins and the same mix of fates, whichever outflow it takes. At the edge of a contributing area, the water of a cell then splits between the sink and other outlets, where particles from the same cell would mostly either reach the sink or miss it. For this reason, areas traced under full mixing are wider and more diffuse than particle areas. Larger cells widen them further (Section 6).

The code reads the cell-to-cell flows (FLOW-JA-FACE; Langevin et al. 2017) and the source and sink terms from the MODFLOW 6 budget file. The flow from cell i to its neighbour j is $q_{ij} \geq 0$. Source terms, such as recharge or leakage from a river, bring water into a cell. Sink terms, such as wells or drains, take it out. Each term is kept separate by package and by boundary, so that a trace can follow the water of individual boundary conditions. A stream reach, a lake or a multi-aquifer well can stay a source or sink term of the aquifer cells it exchanges water with. It can also be traced as a cell of its own, from its package budget file, so that its flows to the aquifer, to other reaches and through the water mover (MVR) become flows q_{ij} like any other. The cells of models linked by exchanges are treated as one set of cells. The choice between a term and a cell changes how a trace splits a sink's water by source (example C in Section 4).

If Src_i and Snk_i are the sums of the source and sink terms of cell i, its total inflow, total outflow and cell budget error are

$$I_i = \sum_j q_{ji} + \mathrm{Src}_i, \qquad O_i = \sum_j q_{ij} + \mathrm{Snk}_i, \qquad r_i = I_i - O_i. \quad (1)$$

The cell budgets of a MODFLOW model do not balance exactly, because the solver stops once the heads and residuals meet convergence tolerances. The cell budget error r_i is usually small but not zero. The code reports this error but never forces it to zero. Because F is written with each cell's outflow and G with its inflow, the error enters the water-budget checks that link them (below).

The backward fraction F. The backward fraction answers the question of where the water of a sink comes from. We refer to the sink terms that a trace follows as the destination. The destination can be one well, one stream reach or a group of wells traced together. The backward fraction F_i is the share of the water leaving cell i that later leaves the model through the destination. Let d_i be the outflow of cell i through the destination, and $P_{ij} = q_{ij}/O_i$ the share of its outflow that goes to cell j. Under full mixing, the water that cell i sends to cell j shares the fate of all the water leaving j. In volumes and in shares, this gives

$$F_i\, O_i = \sum_j q_{ij}\, F_j + d_i, \qquad F_i = \sum_j P_{ij}\, F_j + \frac{d_i}{O_i}. \quad (2)$$

A cell's F is the share of its outflow that leaves through the destination directly, plus the outflow-weighted sum of the F of its downstream neighbours. Eq. (2) holds in every cell, so one solve of these linear equations gives the fractions to every destination group at once. Where water circulates among cells, as it can with the XT3D option of MODFLOW 6 for anisotropic flow (Provost et al. 2017), the code solves each such flow loop exactly. FlowSource instead deletes the smallest flow in each loop (Foley and Black 2013; Supporting Information, Text S1).

A map of F shows where a destination's water comes from. A contributing area is the set of cells where F exceeds a threshold that the user chooses, such as 0.5. Because F takes in-between values at the edge of an area, the choice of threshold is a judgement (Section 6). The sum of F times the cell area gives a size of the area without a threshold. We refer to it as the expected area.

Fig. 1a works through an eight-cell example in which water enters at the west edge and as recharge and leaves through a well and a river. The well cell receives 11 m^3/d, of which its well pumps 6 m^3/d. The other 5 m^3/d flow on to the river without returning to the well. Eq. (2) then gives F = 6/11 = 0.55 in the well cell. All the water of the two cells west of the well cell flows into it, so they share its F. The cells of the upper row send part of their water past the well to the river, so their F is lower.

Gross and net accounting at weak sinks. The well cell of Fig. 1a is a weak sink, because its well takes only part of the cell's outflow (Section 1). At a weak sink the method needs a rule for how much of the cell's water to attribute to the sink. Eq. (2) applies gross accounting (Section 1). Each sink term takes its share of the cell's total outflow. Only that share counts as the sink's water, while the rest of the cell's outflow passes on to its neighbours. Gross accounting is the default because it attributes to each sink only the water that the sink takes from the model, as the MODFLOW 6 Groundwater Transport (GWT) Model does (Langevin et al. 2022), so that the water traced to a well adds up to what it pumps.

Under net accounting, the destination cells are fixed at F = 1. All the water that flows into them from other cells then counts as the destination's water. This is how FlowSource attributes water at a well (Foley and Black 2013; Supporting Information, Text S1). In Fig. 1a, net accounting adds the 5 m^3/d that flow on through the well cell to the river. It leaves out the 1 m^3/d of recharge on the well cell itself, because it works from the cell-to-cell flows only. The well is therefore attributed the 10 m^3/d that flow into its cell from its neighbours, against the 6 m^3/d it pumps. The difference is largest at a small well in a cell that carries much more water than the well pumps (Table 1, row 02).

The forward fraction G. The forward fraction answers the opposite question, which is where the water of a source goes. The forward fraction G_i is the share of the water entering cell i

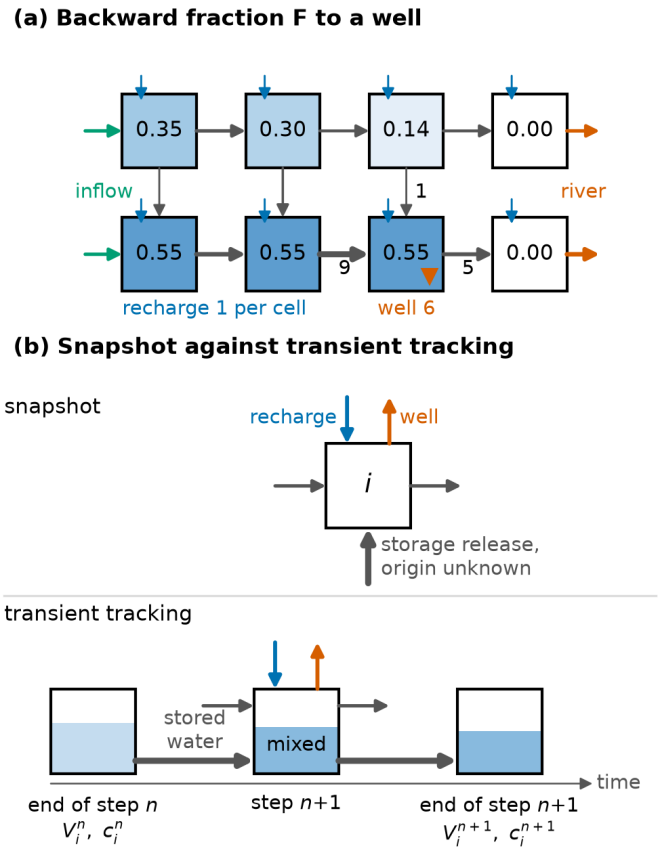


FIGURE 1 (a) Backward fractions F (in cells) to a 6 m^3/d well in an eight-cell example (gross accounting). Water enters from the west and as recharge of 1 m^3/d per cell. It leaves through the well and a river. Arrow widths show flows. Only the well cell's flows are numbered (m^3/d). In the well cell, F = 6/11 = 0.55. (b) One cell i of a transient model. A snapshot (top) traces each time step on its own flows. Water released from storage then enters the cell as a source of unknown origin. Transient tracking (bottom) carries the water stored in the cell at the end of step n into step n + 1. There it mixes with the inflows of that step. V is the cell's water volume and c the share of its water that is tagged. The height of the fill shows V and its shade shows c.

that came from the source. Let s_i be the inflow of cell i through the source. The volume of water entering cell i that came from the source is then

$$G_i I_i = \sum_j q_{ji} G_j + s_i. \quad (3)$$

A cell's G is the share of its inflow that enters directly from the source, plus the inflow-weighted sum of the G of its upstream neighbours. Eq. (3) mirrors Eq. (2), with inflows in place of outflows. A map of G shows how much of each cell's water came from the source (Fig. 2c). G equals the concentration that GWT reaches at equilibrium on the same grid, with

upstream weighting and no dispersion, when the source water carries a concentration of 1 and all other water a concentration of 0 (Table 1, row 03).

Volumes and the water-budget checks. A fraction says what share of a cell's water reaches the destination, but not how much water that is. A fraction times a flow gives a volume of water per unit time. We refer to these rates as volumes. The volume through cell i is $F_i O_i$. It says how much of the water passing through the cell later reaches the destination. The volume from cell i is F_i times the cell's inflow through its source terms. It says how much water enters the model in that cell and later reaches the destination. The volumes from all cells, added up package by package, split the destination's water by source (Fig. 2b). In the same way, the volumes of a forward trace split a source's water by the sink through which it leaves.

Because they are volumes of water, the results of a backward or forward trace can be checked against the flows that MODFLOW 6 simulated. The closure check follows from adding Eq. (2) over all cells. It says that the source water traced to the destination, once corrected for the cell budget errors, adds up to the water the destination takes. The forward–backward check requires the volume from a source to a destination to be the same whether it is traced backward from the destination or forward from the source. The boundary-control check traces every sink of the model at once. It requires the fractions to all sinks to add up to 1 in every cell with outflow, so that the water of every cell leaves the model somewhere. All three checks hold exactly for any flow field, except in loops where water circulates without ever reaching a sink (Supporting Information, Text S1). As a result, a check error larger than round-off points to a fault in the tracing. The checks test the solution of the equations. They cannot show that the flow model is right, or that full mixing describes how water moves in the aquifer.

Mean travel time. Fractions say how much water reaches a destination, but not when it arrives. Travel times, unlike fractions, need a porosity. The water volume V_i of cell i is porosity times the saturated volume of the cell. Water stays in a fully mixed cell for a mean residence time $\tau_i = V_i/O_i$, whichever outflow it then takes. Let t_i be the mean travel time from entering cell i to leaving the model through the destination, for the water that does leave there. The code solves for $m_i = F_i\, t_i$, because m_i obeys an equation of the same form as Eq. (2),

$$m_i = \tau_i F_i + \sum_j P_{ij}\, m_j. \tag{4}$$

Each cell adds its own residence time, weighted by its F, to the times of its downstream neighbours. The code reports the mean travel time $t_i = m_i/F_i$ only where F is above a small threshold (Fig. 2d), because where F is close to zero the mean rests on very little water. Mean travel times are defined for gross accounting only. Eq. (4) weights each cell's residence time by the share of its outflow that leaves through the destination. Under net accounting a destination cell has F = 1, although part of its water flows on to other sinks, so this weight no longer matches the water that leaves through the destination. A mean travel time averages fast and slow water that have mixed in the cells, so it is not a first-arrival time.

Transient flow: snapshots and transient tracking. FlowSource (Foley and Black 2013) traces a transient model one time step at a time, as if the flows of each step were steady-state (Environment Agency 2019). We refer to such a trace as a snapshot. Water released from storage then counts as a source term. Water taken into storage counts as a sink term (Fig. 1b, top). This is a limitation, because in a transient model the water that reaches a sink in one step may have entered the aquifer in an earlier one. A snapshot cannot show recharge of the wet season reaching a well in the dry season. It also cannot say where the water released from storage came from.

We extend full mixing through time. We refer to this as transient tracking. Water from the chosen sources is tagged, as a tracer would be, and carried from step to step through the water volume of each cell (Fig. 1b, bottom). The hydraulic mixing-cell method of Partington et al. (2011) carries mixed water through storage in a similar way in integrated surface water–groundwater models. Storage terms are not traced as source or sink terms. Instead they change the water volume of a cell over step n from V_i^{old} to V_i^{new}, with $V_i^{\text{old}} - V_i^{\text{new}} = \Delta t^n S_i$, where Δt^n is the length of the step and S_i the net release from storage. Let c_i^n be the share of the water in cell i at the end of step n that is tagged. With all flows taken from step n,

$$V_i^{\text{new}} c_i^n - V_i^{\text{old}} c_i^{n-1} = \Delta t^n \Big[\sum_j q_{ji}\, c_j^n + s_i^n - (O_i + r_i)\, c_i^n \Big]. \tag{5}$$

Here O_i and s_i leave out the storage terms, but r_i is still the cell budget error of Eq. (1), with the storage terms included. In Fig. 1b (bottom), V_i^n stands for V_i^{new} of step n.

Eq. (5) is the balance of tagged water in one cell over one step. The outflows carry the cell's mix at the end of the step, because the scheme is implicit in time, as in GWT (Langevin et al. 2022). The term $r_i c_i$ lets the cell budget error leave the cell at the cell's own mix, as the flow-imbalance correction of GWT does. The code applies this correction by default. Water moves by advection only, with no dispersion, diffusion, sorption or decay. Some spreading still comes from mixing whole cells and from long time steps (Supporting Information, Text S1).

Transient tracking needs a porosity and the budget of every time step. The Output Control setting SAVE BUDGET ALL writes this budget. We refer to the water in the model at the start of the simulation as initial water. It can be tagged as its own group. Two choices also affect the result. By default, the water volume of each cell starts at porosity times saturated volume and then changes only by the storage flows, so that every cell's

water balance closes exactly. As an option, the volume is reset to porosity times saturated volume at the end of each step, as in GWT, so that results can be compared with a GWT run. Long implicit time steps make tagged water arrive too early. For this reason, each flow step can also be split into shorter transport sub-steps without a change to the flow model. The SI describes both options (Supporting Information, Text S1). Example B (Section 4) shows their effect.

Backward tracking in time. Forward tracking says how much of each cell's water came from a source. It cannot say which of the water now in the aquifer will reach a sink, or by when. Backward tracking in time answers this question. It solves a balance of the same form as Eq. (5), with the roles of inflow and outflow swapped, from the last step of the simulation back to the first. This differs from MODPATH backward tracking, in which particles start at the sink and move against the flow. Backward tracking in time is the discrete counterpart of the adjoint methods that solve a transport equation backward in time (Neupauer and Wilson 1999, Cornaton and Perrochet 2006).

The backward fraction in time w_i^n is the share of the water in cell i at the start of step n, or entering it during the step, that leaves through the destination before the simulation ends. Over step n,

$$\left[V_i^{\text{new}} + \Delta t^n\,(O_i + r_i)\right] w_i^n = \Delta t^n \left[\sum_j q_{ij}\, w_j^n + d_i^n\right] + V_i^{\text{old},\,n+1}\, w_i^{n+1}. \tag{6}$$

The water in a cell during a step can reach the destination by three routes. It can leave through the destination directly, flow to a neighbour, or stay in the cell into the next step, with the volume at the start of that step. No water can reach the destination after the last step, so the solve starts there with w = 0.

A map of w at the start of the simulation shows the share of each cell's water that the destination takes before the simulation ends, for example within 30 years (Fig. 2f). A threshold on this map gives a time-limited contributing area. The same backward run also gives the share of each cell's water that leaves through other sinks and the share still stored in the model at the end. The three shares add up to 1, apart from a remainder that the code reports (Supporting Information, Text S1). As in steady-state flow, a forward–backward check links the forward and backward runs (Supporting Information, Text S1). When the flows are held constant for long enough, c tends to the steady-state forward fraction G. In the same limit, w at the first step tends to the steady-state backward fraction F.

3. SOFTWARE AND COST

The method is implemented in `gwsource`. It is an open-source Python package (MIT licence). It reads the binary grid file and the budget file of a MODFLOW 6 Groundwater Flow (GWF) model, on any grid type. Stream reaches, lakes and multi-aquifer wells can be traced as extra cells from their own budget files. The inputs, selections and outputs are listed in Supporting Information, Table S2.

The regional model of Section 5 has 1.68 million cells. On an Apple M4 Max with 64 GB of memory, one trace of its eight wells, alone and together, took 2–3 s. Its memory peaked at 2.5 GB. The MODFLOW 6 run of that model takes 44 s (Supporting Information, Text S2). The package is tested on ten small test models and two published examples (Supporting Information, Figure S1).

4. WORKED EXAMPLES

This section shows how a practitioner can use the code to answer three common questions and how far each answer can be trusted. The three worked examples ask where a well's water comes from (example A), how much of it is already in the aquifer (example B), and how much comes from each source and how old it is (example C). Each compares its result with another common approach, such as particle tracking, transport with tagged water or a published study. The example sink is a well, but each output applies in the same way to a stream reach, a drain, a lake or a group of them. Fig. 2 shows one output of each type. Table 1, at the end of this section, summarises the ten notebooks that ship with the code.

All examples start from one small example model. It is a single unconfined layer of 40 × 40 cells of 50 m, with recharge of 5×10^{-4} m/d on every cell. The layer is 50 m thick and has a hydraulic conductivity of 5 m/d. A constant head along the west edge brings in regional groundwater. A river runs north–south 100 m east of a supply well that pumps 800 m^3/d. Each example ships with the code as a Jupyter notebook that builds its model, reproduces its numbers and draws its figures (Section 7).

Example A: where does the well's water come from? A protection zone around a supply well starts from the land whose water reaches the well. For this reason, a practitioner needs to know how large that area is, how far it reaches and how much to trust its edge. Fig. 2a maps F for the supply well. The size of the contributing area depends on the threshold that the user chooses. Cells with F > 0.5 cover 1.143 km^2, against 1.575 km^2 for cells with F > 0.1. The expected area of Section 2 does not need a threshold. Here it lies between the two, at 1.173 km^2. Both areas reach the west edge of the model, about 1.4

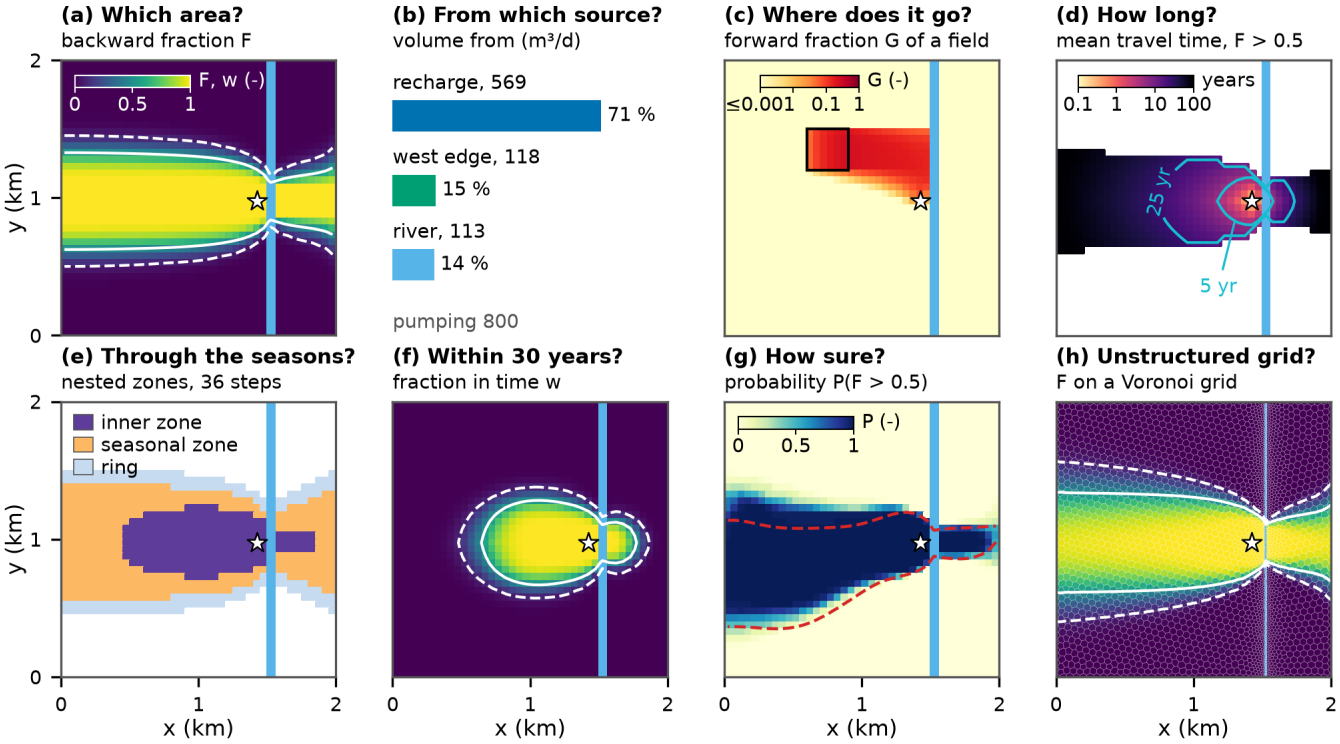


FIGURE 2 Outputs for the example well (Section 4). (a) Backward fraction F. White lines: F = 0.1 (dashed) and 0.5 (solid). (b) The well's water by source (m^3/d). (c) Forward fraction G of recharge on the outlined field. (d) Mean travel time where F > 0.5. (e) Cells with F > 0.5 in all (inner zone) or some (seasonal zone) of 36 seasonal snapshots. Ring: F > 0.1 in some snapshots. (f) Share of each cell's water pumped within 30 years. (g) Share of 50 posterior realisations with F > 0.5. Red dashed line: true F = 0.5. (h) F on a Voronoi grid. Panels (f) and (h) use the colours and lines of (a). Star: well. Blue cells: river.

km from the well. They reach this far because 15 % of the well's water enters at the west edge as regional inflow. Both areas also extend east of the river, as far as the east edge of the model. Near the well the river loses water to the aquifer, so groundwater from the east can pass through the river cells to the well.

The closure check of Section 2 tests the trace behind these numbers. The traced sources add up to the 800 m^3/d pumped, with a relative closure error of 0.0. This check tests the solution of the equations, but it cannot test the position of the area's edge. The edge is where full mixing and particle tracking are expected to differ, so it needs a check of its own. For this check the code runs MODFLOW 6 PRT on the same flows. Particles start from every source term in cells with F > 10^{-6}. Each particle carries a share of its term's flow as a weight. It loses part of that weight at every other sink it passes, in proportion to the flow that sink takes from the cell. We refer to these as weight-losing particles. They are the particle counterpart of gross accounting. Similar weighting rules at sinks appear in the travel-time method of Visser et al. (2009) and the solute-transport method of Winston et al. (2018). Because the particles lose weight at each sink by the full-mixing shares, they are not an independent particle reference. They show how much full mixing spreads water between sinks.

A particle run conserves water when the weight its particles deliver to each sink equals the water that sink takes. Weight-losing particles conserve water, so they deliver what the well pumps whatever the shape of the area (Table 1, row 01). The comparison of areas is more telling. On this model the F > 0.5 area is close in size to the area from which the particles deliver most of their water. The particle share of a cell is the share of the weight released there that reaches the well (Fig. 3a). We refer to the cells with a particle share above 0.5 as the particle area. It covers 1.168 km^2. The F > 0.5 area is 2.1 % smaller and leaves out 14 of its cells. The F > 0.1 area is 35 % larger. It holds every cell of the particle area. The two methods differ most at the edge of the area. In the 372 cells with F between 0.05 and 0.95, the particle share and F differ by 0.164 on average. In those cells most particles either reach the well or miss it, while full mixing gives the cell an in-between value (Section 2).

For the practitioner, this means that on this model the F > 0.5 area is a fair estimate of the particle area. Because the F > 0.1 area holds every particle cell, it serves here as a cautious outer

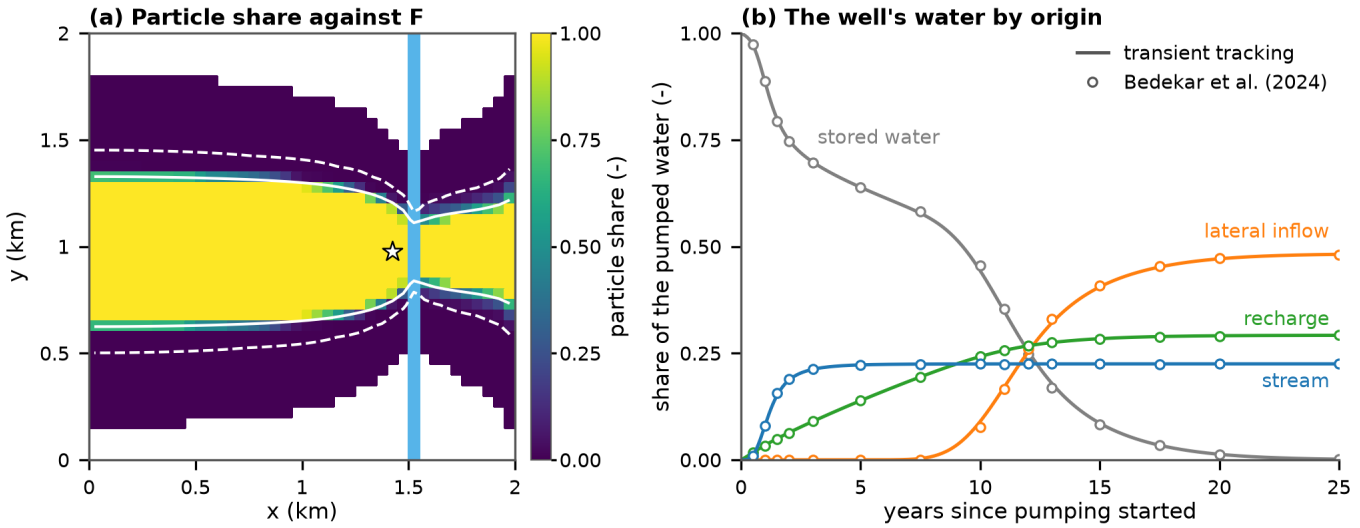


FIGURE 3 Checks of worked examples A and B. (a) Particle share in each cell of the example model where weight-losing PRT particles start. The particle share is the share of the water released in a cell that the particles deliver to the well. Cells without particles are blank. White lines: F = 0.1 (dashed) and F = 0.5 (solid) from full mixing. Star: well. Blue cells: river. (b) The well's water by origin in the unit-concentration example of Bedekar et al. (2024), rebuilt in MODFLOW 6. Stored water is the water in the aquifer at the start of the run, one day before pumping starts. It is the initial water of Section 2. Lines: forward transient tracking. Open circles: values read off Figure 2 of Bedekar et al. (2024).

limit. Section 5 shows that a regional model need not behave in the same way.

Example B: how much of the pumped water is already in the aquifer? A manager may ask how much of the water pumped over the coming decades is already in the aquifer today. The answer matters where new recharge differs in quality from the water already stored, for example after a change in land use. A seasonal version of the example model shows why snapshots cannot answer this question (Section 2). In this version, recharge is multiplied by 1.8, 1.0, 0.1 and 1.1 in the wet, wet-to-dry, dry and dry-to-wet seasons. The well pumps between 640 and 1,120 m^3/d. The model runs for 30 years, with one time step per season and 121 time steps in all. Seasonal snapshots show which cells lie in the contributing area in every season and which in some seasons only (Fig. 2e; Table 1, row 05). Averaged over the 30 years, the snapshots give the well's water as about 60 % recharge and 15 % storage release (Supporting Information, Figure S2a). A snapshot cannot say where that stored water came from. It may be old water or recharge of the last wet season.

Transient tracking follows the stored water. It shows that most of the water pumped in the first decades was already in the aquifer. The tracking uses a porosity of 0.3 and four transport sub-steps per flow step. The run starts from a steady state in which the well already pumps. Initial water is therefore not water from before pumping began. Of all the water pumped in the first 10 years, 81 % was initial water. This share is 98 % for the first year and 68 % over 30 years. Even in the last season of year 30, the well's water is 51.6 % initial water, 32.9 % recharge and 15.5 % river water. In that season recharge supplies about a third of the well's water, against about 60 % averaged over the snapshots. New recharge mixes into a large volume of older water, because a column of this aquifer holds about 64 years of recharge. With one layer, the model cannot hold young water above old water, so a layered model may give other shares.

Backward tracking in time shows which of the water now in the aquifer reaches the well. Only the water near the well and along the main flow path reaches it within 30 years (Fig. 2f). Most of the water in the steady-state contributing area is still on its way at the end. Of the 8.77 million m^3 pumped over 30 years, backward tracking attributes 67.7 % to initial water. This equals the 68 % of the forward run, as the forward–backward check requires.

On the same model, transient tracking with GWT's water volumes and one transport step per flow step matched a GWT run with tagged recharge to 9.8×10^{-14} in every cell and step. Both codes solve the same equations, so this is a consistency test. It shows that the code solves Eq. (5) as GWT does. With one transport step per flow step, the two volume options of Section 2 change the fractions by less than 0.004 on this model, although porosity (0.3) differs from specific yield (0.2) (Supporting Information, Text S3). Transport sub-steps matter more, because long implicit steps make new water arrive too early.

Four sub-steps per flow step remove most of this effect on this model (Supporting Information, Text S3).

The GWT check compares two codes that solve the same equations. An external check needs a published transient study made with other codes. Bedekar et al. (2024) used the unit-concentration method to show how a new well's water changes from stored water to water from its sources. This method is the transport with tagged water of Section 1. Their model has three layers, with lateral inflow along its west edge, recharge on the top layer, a stream along its east edge and a well that pumps for 25 years. They ran MODFLOW-2005 and MT3D-USGS, with one transport run per source. We rebuilt their model in MODFLOW 6 (Supporting Information, Text S2). One forward tracking run with four tags reproduces the published curves (Fig. 3b). At year 25, the well's water is 48.2 % lateral inflow, 29.2 % recharge and 22.5 % stream water. Bedekar et al. (2024) give 48 %, 29 % and 23 %. Over 15 times read off their Figure 2, the largest difference is 1.3 percentage points, on the lateral-inflow front at year 10. The values read off the published figure are themselves good to about one percentage point. Backward tracking in time adds a map of the time-limited contributing area (Supporting Information, Figure S3). Within 25 years, this area comes close to the steady-state F > 0.5 area of the final flows. We expect that a forward unit-concentration model would need one run per zone of the aquifer to give such a map, though we have not tested this.

For the practitioner, this means that snapshots and transient tracking answer different questions. A snapshot treats the flows of each step as steady, so it says which sources feed the well but not when their water entered the aquifer. Transient tracking shows that most of the water pumped over the next decade is already in the aquifer. Where new recharge differs in quality from the stored water, we would therefore expect the well's water to change over decades rather than seasons on a model like this one.

Example C: how much water comes from each source, and how old is it? Many studies need a sink's water split by source more than the edge of its area. A water-quality study, for example, may need to know how much of a well's water is river water and how old the water from each source is. On the example model, the well's total water and the mean travel time of all its water depend much less on mixing than the edge of an area does. Section 5 shows that on a regional model the split by source can still differ between methods by several percentage points. The backward trace of example A splits the 800 m^3/d pumped into 71 % recharge, 15 % inflow at the west edge and 14 % river leakage (Fig. 2b). A forward trace asks the reverse question of where a source's water goes, for example the recharge under a field where fertiliser is spread. Of the recharge on a field of 300 m × 300 m, 39 % reaches the well (Fig. 2c). A forward trace from each source gives the same split of the well's water as the backward trace (Table 1, row 03). In a test problem with a river and recharge, the shares by source were the same to three digits whether water was mixed across whole cells, routed from face to face or followed with particles on the same grid (Supporting Information, Text S5).

The split by source also depends on where the trace stops. A second version of the example model shows this. In it, the river becomes a stream that flows from north to south. Two drains pass their water to the stream through the water mover. The well of this version takes a different mix of water from the well of example A. For this reason, its shares cannot be compared with the 14 % of river leakage given above. When the stream reaches stay source and sink terms of the aquifer cells, stream leakage supplies 32 % of the well's water. When the reaches are traced as cells of their own, that water splits into 129 m^3/d of inflow at the upstream end of the stream and 126 m^3/d of recharge that had discharged to the drains or the stream. A study that reports how much of a well's water is stream water should therefore say which of the two choices it made.

Full mixing and weight-losing particles give nearly the same mean travel time for all of the well's water, but they agree less well for river water. With a porosity of 0.3, the backward trace also gives mean travel times (Fig. 2d). They are taken to the point where the water enters the well cell, because particle trackers stop particles there (Supporting Information, Text S1). The mean travel time is 2.1 years for river leakage, 41.6 years for recharge and 108.4 years for inflow at the west edge. Over all the water it is 45.9 years, against 45.7 years for the particles (−0.5 %). This agreement is not an independent test. Any method that conserves water gives the mean travel time to the well as the volume of water bound for the well divided by its inflow, here 46.0 years. The time to the well cell is shorter by the time the water spends in that cell, about 0.1 years. The agreement therefore tests only the volume attributed to the well. That volume is nearly the same in both methods. By source, the particle times differ by −2.1 % for recharge and +2.1 % for the west edge, but by −20.1 % for river leakage. We attribute the river difference to its short path. River water reaches the well through few cells, so the mixing in those cells weighs more in its mean.

The comparison cell by cell is the informative one. In the 456 cells with F ≥ 0.5, apart from the well cell, the median ratio of particle time to full-mixing time is 0.96. In the 676 cells with F < 0.5, it is 5.72. A particle from a cell with small F follows one slow path around the edge of the area. Full mixing instead averages over all the paths that the cell's water takes. The same difference shows in the time-of-travel zones of Fig. 2d. The 25-year zone covers 0.455 km^2 under full mixing and 0.465 km^2 for particles (+2.2 %). The 5-year zone covers 0.098 against 0.122 km^2 (+25.6 %). On this evidence, we would take a well's overall mean travel time from full mixing. We would

check its split by source with particles where the split matters (Section 5). We would draw a short time-of-travel zone with particles (Section 6).

All ten notebooks. Table 1 lists all ten notebooks. Eight of them ran in 10 s or less. Row 04 runs the trace of example A on other grid types (Fig. 2h). Row 09 applies the trace to an ensemble of models. Many sets of parameters can fit the measured heads about equally well. We refer to each set as a realisation and to a group of realisations as an ensemble. PEST++ IES is the iterative ensemble smoother of the PEST++ software (White 2018, White et al. 2020). History matching with PEST++ IES adjusts the realisations until they fit the measurements. We refer to the ensemble before history matching as the prior and to the history-matched ensemble as the posterior. In row 09, the measurements come from a synthetic true model, so the posterior can be compared with the truth (Fig. 2g; Supporting Information, Figure S4). A narrower posterior is not necessarily a correct one. One synthetic case cannot show how reliable posteriors are in general. Here the trace ran on each realisation after PEST++ IES had finished. We have not tested running it inside each PEST++ model run.

5. A REGIONAL MODEL: NEVERSINK

This section shows that a trace works and stays fast on a published regional model. It also explains why full-mixing areas differ from the published particle areas, so that a practitioner can read one next to the other. The USGS Neversink/Rondout model (Fienen et al. 2022, Corson-Dosch et al. 2022) is a steady-state MODFLOW 6 model of 1.68 million cells. The published study refers to eight of its wells as priority wells. Together they pump 3,671 m^3/d. The model has 358 realisations from history matching with PEST++ IES (Section 4). We refer to the realisation with the published calibrated parameter values as the calibrated model.

The published study mapped the wells' contributing area in the calibrated model with MODPATH 7 (Pollock 2016). One particle started in the uppermost active cell of each column near the wells. We refer to these cells as the start cells. The published contributing cells are the 980 start cells whose particle ends in a priority well. We traced the wells in the calibrated model with gross accounting. The stream reaches stayed source and sink terms of the aquifer cells.

A trace of all the wells took a few seconds, less than the fastest PRT run on the same flows. On an Apple M4 Max with 64 GB of memory, one trace of nine destination groups (each well alone and all eight together) took 2–3 s and peaked at 2.5 GB of memory. Each further group added about 0.19 GB (Supporting Information, Text S4). The closure check held to 1 $\times 10^{-12}$ m^3/d or better for every group. PRT took about 30 s when it wrote only where each particle ends. With full particle tracks, it took about 5 min with particles released at one height in each cell and about 13 min with particles released at four heights. These times do not compare like with like. One trace gives F in every cell, the volumes and the checks. A PRT run gives particle shares only in the cells it seeds. PRT memory was not measured. Across the ensemble, two traces of each realisation, without and with the stream network, took about 9 s.

The full-mixing areas differ from the published area. The overlap of two areas is the number of cells in both divided by the number in either. The published area comes from one reasonable choice of particle settings. The published runs put recharge and stream flows on the top face of a cell where no active cell lies above it. They stopped a particle in the first weak-sink cell it entered. They moved a particle that started in a dry cell down to the first active cell below. We refer to this setting as draping. PRT on the same flows reproduces the published area with these settings (overlap 0.998). Without the top-face setting, the overlap falls to 0.65. Without draping, it falls to 0.96. Two of our particle runs do not conserve water in the sense of Section 4. Particles that lose weight only at the wells deliver 2.6 times the wells' inflow (overlap 0.29). Particles released at the water table deliver 0.66 times it (overlap 0.39). Leaving these two runs out, the particle settings alone move the overlap from 0.62 to 0.998 (Supporting Information, Text S4).

The full-mixing areas overlap the published area less than any of these particle runs do. The areas where F exceeds 0.5 and 0.1 have overlaps of 0.41 and 0.30. We infer that the main reason is the one seen in example A. Full mixing splits each cell's water among all its outflows, while a particle follows one path. On average, the published cells send 42 % of their water to the wells. Many of them therefore fall below F = 0.5 (Fig. 4a). The F > 0.1 area holds 73 % of them (711 of 980). Fig. 4b shows this for a window around four of the wells, where the F > 0.1 area holds 538 of the 688 published cells.

Weight-losing particles (Section 4) lose weight at each sink as full mixing does. For this reason they measure only the spreading that full mixing adds between sinks. The F > 0.1 area holds 87 % of the 691 start cells where these particles deliver more than half their water to the wells (Supporting Information, Figure S5a). This share is an upper bound, because the particles started only in cells with F > 10^{-6}. They therefore needed the trace before they could run. The 89 cells it misses supply 6 % of the particles' water to the wells (227 m^3/d).

Weak sinks add to the difference. A buried stream cell holds a stream reach and has another active cell above it. Where such a reach gains water, it is a weak sink that stops published particles. In the published runs, 26,828 particles stopped in buried stream cells and 466 in the five priority wells that take only part of their cell's outflow. Each stopped particle counts in full as reaching that sink, while full mixing passes the rest

TABLE 1 The ten notebooks that ship with the code. Each row gives a notebook's question, its main output, the other approach or published study it is compared with and how closely the two agree. Letters after the notebook number give the worked example of this section that describes the notebook in detail. Every notebook also runs the water-budget checks of Section 2. Differences of about 10^{-10} or smaller are round-off. All notebooks use the example model of this section or a version of it (Supporting Information, Text S2). Supporting Information, Text S3 describes the rows that the worked examples do not cover in full.

Notebook	Question	Output	Compared with	Agreement	Fig. 2 panel
01 (A, C)	Where does a well's water come from?	F map, F > 0.5 and F > 0.1 areas, the well's water by source	Weight-losing PRT particles	F > 0.5 area 2.1 % smaller than the particle area	a, b
02	What is computed, and what changes at a weak sink?	Gross and net accounting, fractions to every sink. At a small well that pumps 5 m^3/d from a cell that carries 33.7 m^3/d, net accounting attributes all 33.7 m^3/d to the well (6.7 times its pumping)	The published worked example of Foley and Black (2013)	Published example reproduced to 1×10^{-14}. Fractions to all sinks add up to 1 within 1×10^{-15}	–
03 (A, C)	Where does a source's water go?	G maps, fate of the recharge on a field	GWT with tagged water, run to equilibrium	4.2×10^{-13} in all 1,600 cells	c
04	Does it work on my grid?	The same trace on regular (DIS), Voronoi (DISV) and fully unstructured (DISU) grids. Flow loops with XT3D	The same model on the regular grid. A solve of the whole system at once	DISU equals DIS in every cell. On DISV the expected area is 1.178 against 1.173 km^2 (+0.4 %), but F falls from 0.5 to 0.1 over 186 m against 121 m. The loop-by-loop solve agrees with the whole-system solve to 3.3×10^{-16}	h
05 (B)	How does the area change through the seasons?	Seasonal snapshots of a 3-year run. The F > 0.5 area grows from 0.59 km^2 in the wet season to 1.35 km^2 in the dry season	–	Every snapshot closes within 1×10^{-6} m^3/d	e
06 (C)	How old is the water reaching the well?	Mean travel time map, 5- and 25-year zones, travel time by source	Weight-losing PRT particles	All water within 0.5 %, not an independent test. Wide differences at the edge of the area	d
07 (C)	What if water passes through streams, drains, wells or linked models?	The well's water by source, with stream reaches traced as cells of their own or not. A multi-aquifer well. A model split into three linked models	The same model unsplit	2.1×10^{-13} in F	–
08 (B)	How much of the pumped water was already in the aquifer?	The well's water by origin through time. Backward fractions in time	GWT with tagged recharge	9.8×10^{-14} over 1,600 cells and 121 steps	f
09	How does parameter uncertainty change the area?	Probability of contribution over 50 realisations, before and after history matching with PEST++ IES	A synthetic true model	The band with a probability between 0.1 and 0.9 shrinks from 1.10 to 0.35 km^2. A probability of at least 0.5 covers 93 % of the true F > 0.5 area (prior 79 %). Of the cells with a probability of at least 0.9, 86–87 % lie in the true area, before and after. Histogram quantiles within one bin width of the exact ones. Serial and parallel runs identical	g
10 (B)	Does it reproduce a published unit-concentration study?	The well's water by source over 25 years. Time-limited contributing areas	Bedekar et al. (2024), Figure 2. Four GWT runs	Within 1.3 percentage points of the published curves (0.6 up to year 15 with 7.6-day sub-steps). GWT to 1.5×10^{-10}	–

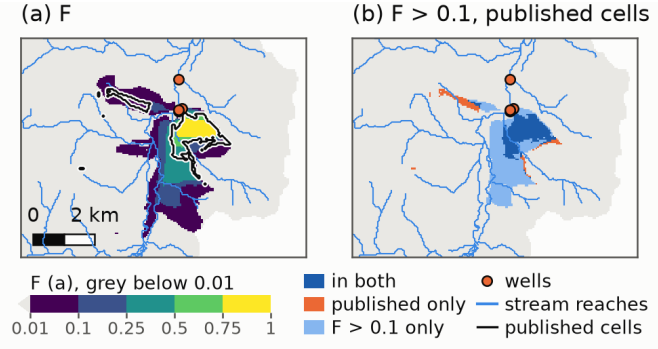


FIGURE 4 F and the published contributing cells of the Neversink model, in a 7.1 × 9.25 km window around four of the priority wells. The window holds 688 of the 980 published cells. (a) F in the calibrated model, with the published cells outlined. (b) The F > 0.1 area against the published cells. Dark blue: in both (538 cells). Orange: published only (150). Light blue: F > 0.1 only (1,090).

of the cell's water on (Supporting Information, Text S4). Net accounting attributes to a weak-sink well all the water that enters its cell. It raises the F > 0.5 overlap only from 0.41 to 0.44. At the weak-sink wells, gross accounting comes closer to the published cell counts than net accounting does.

Example C (Section 4) showed that a sink's split by source depends on how stream reaches are traced. On this model the split also depends on the method. With the stream reaches left as source and sink terms, full mixing attributes 57 % of the wells' water to recharge and 43 % to stream leakage. With the reaches traced as cells of their own, all of the wells' water traces back to recharge. Along weight-losing particle paths, with the reaches again left as source and sink terms, the split is 49 % recharge and 51 % stream leakage. Both methods deliver the wells' total inflow to within 2 %. We infer, without tracing it cell by cell, that stream water leaking near the wells stays on particle paths that end at a well. Full mixing instead gives that water the average fate of all the water leaving the cells it passes. A split by source on a regional model should therefore be reported with the method that gave it.

For the practitioner, a particle area depends on its particle settings as much as on the flow model. The comparison that isolates full mixing is a particle run that loses weight at sinks as full mixing does. Against that run, the F > 0.1 area held 87 % of the particle area on this model, so a low threshold on F did not give a complete outer limit.

6. TRADE-OFFS AND CONCLUSIONS

Full-mixing areas are wider and more diffuse than particle areas (Section 5). Their edge is a band of cells where F lies between 0.1 and 0.9. In two simple test problems, halving the cell size made the band only about 30 % narrower (Supporting Information, Text S5; Supporting Information, Figure S6). On squares turned by 45° to the flow, the band was 1.8 times as wide as on squares aligned with it (Supporting Information, Text S5). Where the edge of an area matters, finer cells along it and a grid aligned with the main flow narrow the band.

No single threshold on F therefore reproduces a particle area everywhere. The threshold is a judgement about the purpose of the area. Both the threshold and the accounting should be reported with the area. In the test problems, on cells of 50 m or less, the F > 0.5 area was within 5 % of the size of the reference area. The F > 0.1 area missed at most 0.1 % of the reference area. It was 8 to 90 % larger. On Neversink, the F > 0.1 area still missed cells that weight-losing particles assign to the wells (Section 5). So a low threshold does not guarantee a cautious area on a regional model.

Transient tracking has so far been tested on synthetic models only. Its memory grows with the number of time steps. Movers between models (the MVE package) are not implemented yet. Without cell vertices, the code cannot map results on a DISU grid. A well's water by source and the mean travel time of all its water depend much less on mixing than the edge of its area does (Section 4). We would therefore use particles where a sharp edge or a first arrival matters, such as a short time-of-travel zone. We would use GWT where dispersion, sorption or decay matter, because transient tracking moves water by advection only (Langevin et al. 2022). Full mixing suits questions about many sinks, unstructured grids, package networks or many realisations.

7. CODE AND DATA AVAILABILITY

The code, `gwsource`, is open source under the MIT licence at https://github.com/rhugman/gw-source [check: public URL]. This paper uses version 0.0.1 [check: release version and tag]. Each example ships with the code as a Jupyter notebook that reproduces its numbers and figure (Supporting Information, Text S2). The same release holds the verification tables (docs/verification.md), the test models (tests/models/README.md) and the notes of the Neversink study (docs/neversink_study.md). The Neversink model is in a USGS data release (Fienen and Corson-Dosch 2021). The files behind every figure will be archived on Zenodo [check: DOI].

REFERENCES

Bedekar, V., Neville, C., Tonkin, M.J., Bartlett, R.D. & Plato, P. (2024) A unit-concentration method to quantify source contribution. *Groundwater*, 62(2), 303–309. doi:10.1111/gwat.13333.

Cornaton, F. & Perrochet, P. (2006) Groundwater age, life expectancy and transit time distributions in advective–dispersive systems: 1. Generalized reservoir theory. *Advances in Water Resources*, 29(9), 1267–1291.

doi:10.1016/j.advwatres.2005.10.009.

Corson-Dosch, N.T., Fienen, M.N., Finkelstein, J.S., Leaf, A.T., White, J.T., Woda, J. et al. (2022) *Areas contributing recharge to priority wells in valley-fill aquifers in the Neversink River and Rondout Creek drainage basins, New York*. U.S. Geological Survey, Reston, Virginia. Scientific Investigations Report 2021–5112.
URL https://doi.org/10.3133/sir20215112

Environment Agency (2019) *Manual for the Production of Groundwater Source Protection Zones*. Bristol, UK: Environment Agency.
URL https://assets.publishing.service.gov.uk/media/5d41a020e5274a0a0bf7757c/Manual-for-the-production-of-Groundwater-Source-Protection-Zones.pdf

Fienen, M.N. & Corson-Dosch, N.T. (2021) *Groundwater model archive and workflow for Neversink/Rondout Basin, New York, source water delineation*. U.S. Geological Survey data release.
URL https://doi.org/10.5066/P9HWSOHP

Fienen, M.N., Corson-Dosch, N.T., White, J.T., Leaf, A.T. & Hunt, R.J. (2022) Risk-based wellhead protection decision support: A repeatable workflow approach. *Groundwater*, 60(1), 71–86. doi:10.1111/gwat.13129.

Foley, C. & Black, A. Efficiently delineating volumetric capture areas and flow pathways using directed acyclic graphs and MODFLOW; description of the algorithms within FlowSource. In: *Proceedings of MODFLOW and More 2013: Translating Science into Practice, Golden, Colorado, June 2–5, 2013*. Golden, Colorado: IGWMC, Colorado School of Mines, pp. 571–581, [check: pages against the proceedings].
URL https://groundwater-science.co.uk/wp-content/uploads/2013_FSv6_DAGPaper.pdf

Goode, D.J. (1996) Direct simulation of groundwater age. *Water Resources Research*, 32(2), 289–296. doi:10.1029/95WR03401.

Langevin, C.D., Hughes, J.D., Banta, E.R., Niswonger, R.G., Panday, S. & Provost, A.M. (2017) *Documentation for the MODFLOW 6 Groundwater Flow Model*. U.S. Geological Survey, Reston, Virginia. Techniques and Methods book 6, chap. A55.
URL https://doi.org/10.3133/tm6A55

Langevin, C.D., Hughes, J.D., Provost, A.M., Russcher, M.J., Reno, M.J., Bonelli, W.P. et al. (2026) *MODFLOW 6 Modular Hydrologic Model version 6.8.1*. U.S. Geological Survey Software Release, 23 September 2026.
URL https://doi.org/10.5066/P1PGE9XW

Langevin, C.D., Provost, A.M., Panday, S. & Hughes, J.D. (2022) *Documentation for the MODFLOW 6 Groundwater Transport Model*. U.S. Geological Survey, Reston, Virginia. Techniques and Methods book 6, chap. A61.
URL https://doi.org/10.3133/tm6A61

Neupauer, R.M. & Wilson, J.L. (1999) Adjoint method for obtaining backward-in-time location and travel time probabilities of a conservative groundwater contaminant. *Water Resources Research*, 35(11), 3389–3398. doi:10.1029/1999WR900190.

Partington, D., Brunner, P., Simmons, C.T., Therrien, R., Werner, A.D., Dandy, G.C. et al. (2011) A hydraulic mixing-cell method to quantify the groundwater component of streamflow within spatially distributed fully integrated surface water–groundwater flow models. *Environmental Modelling & Software*, 26(7), 886–898. doi:10.1016/j.envsoft.2011.02.007.

Pollock, D.W. (2016) *User guide for MODPATH Version 7—A particle-tracking model for MODFLOW*. U.S. Geological Survey, Reston, Virginia. Open-File Report 2016–1086.
URL https://doi.org/10.3133/ofr20161086

Potter, S.T., Moreno-Barbero, E. & Divine, C.E. (2008) MODALL: A practical tool for designing and optimizing capture systems. *Ground Water*, 46(2), 335–340. doi:10.1111/j.1745-6584.2007.00410.x.

Provost, A.M., Langevin, C.D. & Hughes, J.D. (2017) *Documentation for the "XT3D" option in the Node Property Flow (NPF) Package of MODFLOW 6*. U.S. Geological Survey, Reston, Virginia. Techniques and Methods book 6, chap. A56.
URL https://doi.org/10.3133/tm6A56

Visser, A., Heerdink, R., Broers, H.P. & Bierkens, M.F.P. (2009) Travel time distributions derived from particle tracking in models containing weak sinks. *Ground Water*, 47(2), 237–245. doi:10.1111/j.1745-6584.2008.00542.x.

White, J.T. (2018) A model-independent iterative ensemble smoother for efficient history-matching and uncertainty quantification in very high dimensions. *Environmental Modelling & Software*, 109, 191–201. doi:10.1016/j.envsoft.2018.06.009.

White, J.T., Hunt, R.J., Fienen, M.N. & Doherty, J.E. (2020) *Approaches to highly parameterized inversion: PEST++ Version 5, a software suite for parameter estimation, uncertainty analysis, management optimization and sensitivity analysis*. U.S. Geological Survey, Reston, Virginia. Techniques and Methods 7C26.
URL https://doi.org/10.3133/tm7C26

Winston, R.B., Konikow, L.F. & Hornberger, G.Z. (2018) *Volume-weighted particle-tracking method for solute-transport modeling; Implementation in MODFLOW–GWT*. U.S. Geological Survey, Reston, Virginia. Techniques and Methods book 6, chap. A58.
URL https://doi.org/10.3133/tm6A58